

JOHNSON COUNTY EXECUTIVE, KS, USA

WMO: 724468

Lat: 38.850N

Lon: 94.739W

Elev: 1070

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	9.2	-5.1	4.4	6.9	-0.7	5.5	11.5	25.0	36.2	23.1	38.2	8.5	330	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	95.3	75.8	91.9	75.5	89.7	75.1	78.9	89.4	77.7	88.2	76.5	86.3	9.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	139.7	85.1	74.6	134.9	84.0	73.2	128.4	82.4	43.2	89.5	41.9	88.0	40.6	86.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.0	18.1	DB	-2.9	99.5	4.5	3.9	-6.1	102.3	-8.8	104.6	-11.3	106.8	-14.6	109.6
			WB	-3.6	81.2	4.3	1.1	-6.7	82.0	-9.2	82.7	-11.7	83.3	-14.8	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	30.6	34.6	45.3	55.5	65.0	74.3	78.7	77.2	69.3	57.2	44.8	33.8
	DBStd	19.07	11.74	11.43	11.10	9.26	7.69	5.71	5.38	5.62	7.17	9.00	10.08	10.75
	HDD50	2079	606	443	220	50	3	0	0	0	34	214	509	
	HDD65	4822	1066	850	613	305	98	6	0	1	37	272	608	966
	CDD50	4140	5	13	76	214	468	728	890	844	580	257	57	8
	CDD65	1405	0	0	3	18	98	284	425	380	167	29	1	0
	CDH74	13148	0	1	21	142	733	2531	4480	3673	1355	208	4	0
	CDH80	5162	0	0	1	22	168	896	1989	1591	452	43	0	0
(14) Wind	WSAvg	8.8	9.4	9.6	10.4	10.5	8.9	8.4	7.2	7.0	7.5	8.6	9.3	9.2
(15) Precipitation	PrecAvg	39.8	1.3	1.3	2.6	3.7	5.2	5.7	4.0	3.9	4.8	3.3	2.6	1.5
	PrecMax	64.2	3.3	3.9	10.4	10.1	12.4	13.4	15.6	9.1	14.6	8.5	8.5	4.9
	PrecMin	22.8	0.0	0.0	0.6	0.5	0.9	1.2	0.3	0.4	0.5	0.2	0.0	0.0
	PrecStd	8.9	0.9	0.9	1.8	2.0	2.6	3.1	2.9	2.3	3.7	2.2	1.9	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	69.5	77.0	82.7	88.0	94.5	99.6	99.7	93.4	84.4	73.2	64.9
		MCWB	53.2	54.8	62.0	65.7	71.2	75.9	75.4	74.2	72.7	69.8	61.0	58.6
	2%	DB	58.0	63.0	72.1	78.4	84.2	90.5	96.1	95.0	88.6	79.7	69.5	59.8
		MCWB	49.4	52.8	59.5	63.9	69.6	75.2	75.8	75.9	72.9	66.1	59.0	53.4
	5%	DB	53.0	57.0	67.6	74.3	81.4	88.1	92.6	91.3	84.8	75.0	64.2	54.1
		MCWB	45.2	48.3	57.1	61.8	68.6	74.4	76.3	74.9	72.2	64.2	55.2	47.7
	10%	DB	47.0	52.1	63.3	71.0	78.5	85.2	90.0	88.4	81.7	71.9	60.5	49.3
		MCWB	40.6	45.0	54.2	60.4	67.2	72.9	76.2	74.4	69.9	63.0	52.9	43.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.1	59.5	65.3	69.1	74.9	79.1	81.1	80.2	77.1	72.6	64.2	60.2
		MCDB	60.9	65.3	74.0	79.9	84.4	89.2	91.1	91.0	86.4	80.3	69.7	63.6
	2%	WB	50.5	54.3	62.3	66.5	72.5	77.6	79.4	78.3	75.1	69.8	60.2	54.5
		MCDB	56.6	60.8	69.0	74.4	81.0	87.6	89.9	89.3	84.2	76.2	66.2	58.4
	5%	WB	45.3	49.6	59.1	64.2	70.5	75.8	78.2	77.0	73.5	67.1	57.3	48.6
		MCDB	51.4	56.2	65.8	71.6	78.5	84.8	88.7	87.7	82.8	72.7	62.5	52.7
	10%	WB	41.0	44.7	54.6	61.6	68.5	74.4	77.0	75.5	71.8	63.8	53.8	44.0
		MCDB	47.6	51.6	61.9	69.6	76.1	83.0	87.1	85.2	79.7	68.8	59.5	48.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	17.7	19.0	19.0	18.1	17.5	17.9	18.3	18.9	18.8	17.7	16.4
		MCDBR	24.6	26.1	25.0	23.3	20.8	19.8	21.2	22.3	21.3	22.2	22.6	22.6
	5% WB	MCWBR	16.9	17.4	14.9	13.8	10.7	8.9	7.6	8.4	9.6	11.8	14.6	16.4
		MCWBR	22.4	23.6	20.3	20.0	18.7	18.2	18.5	19.5	18.5	18.4	19.6	20.6
(40) Clear-Sky Solar Irradiance	taub		0.290	0.303	0.333	0.381	0.423	0.425	0.434	0.434	0.385	0.341	0.302	0.287
		taud	2.478	2.452	2.424	2.304	2.245	2.270	2.271	2.273	2.364	2.474	2.527	2.513
	Ebn at Noon		283	294	294	284	273	271	268	265	272	276	277	277
		Edh at Noon	25	29	34	40	43	43	42	41	35	29	24	23
(44) All-Sky Solar Radiation	RadAvg		709	951	1236	1573	1807	2053	2044	1813	1521	1089	770	612
	RadStd		74	108	105	100	161	153	128	99	102	123	76	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page